

EE_8 -INVOLUTIONS AND ORDER-3 ISOMETRIES OF THE LEECH LATTICE, WITH MACHINE-CHECKED CERTIFICATES

ABSTRACT. Let Λ be the Leech lattice. An EE_8 -involution of $\text{Aut}(\Lambda) = \text{Co}_0$ is an involution whose (-1) -eigenlattice is a primitive $\sqrt{2}E_8$ sublattice. We exhibit two order-3 isometries of Λ with opposite behaviour: one, with fixed lattice of rank 8 and Gram one third an even unimodular form, commutes with no EE_8 -involution; the other, fixed-point-free of trace -12 , commutes with an explicit one, yielding a fixed-point-free order-6 isometry g with characteristic polynomial $\Phi_6^4\Phi_3^8$. The asymmetry is not accidental. The construction that supplies a commuting EE_8 -involution to *every* monomial isometry of order 2 or 4 proceeds by stabilising a Golay octad, and a fixed-point-free order-3 permutation of the 24 coordinates stabilises none, because $3 \nmid 8$. The finite verifications are formalised in Lean 4: six files, 108 theorems, no `sorry`. An explicit unimodular isometry between the binary-Golay and ternary-Golay models of Λ removes any appeal to Conway’s uniqueness theorem.

1. INTRODUCTION

Write Λ for the Leech lattice, normalised so that its minimum is 4, and $\text{Co}_0 = \text{Aut}(\Lambda)$. Following Griess and Lam [GL1, GL2], call a primitive sublattice $N \subset \Lambda$ isometric to $\sqrt{2}E_8$ an EE_8 -sublattice. Since the discriminant group N^*/N is 2-torsion, the map which is -1 on N and $+1$ on $\text{ann}_\Lambda(N)$ is an isometry of Λ ; we call it the associated EE_8 -involution. These involutions are the lattice shadow of the $2A$ involutions of the Monster, and the pairs of them generate the dihedral configurations at the heart of the moonshine path from E_8 to M .

This note concerns a single question: *which isometries of Λ commute with an EE_8 -involution?* The answer turns out to be governed by an elementary divisibility fact, and the two halves of the answer at order 3 are genuinely different in character — one is an impossibility proved by exhaustion, the other an explicit construction.

Throughout, “ $\sqrt{2}E_8$ ” abbreviates the certified statement “an even lattice of rank 8 whose Gram matrix, divided by 2, is even and unimodular”; the identification of that genus with E_8 is classical and is one of five inputs we do not formalise (§7).

2. THE REDUCTION

Proposition 1. *The assignment $\varepsilon \mapsto \Lambda_-(\varepsilon)$ is a bijection between EE_8 -involutions of Co_0 and EE_8 -sublattices of Λ . For $h \in \text{Co}_0$,*

$$[\varepsilon, h] = 1 \iff \Lambda_-(\varepsilon) \text{ is } h\text{-invariant.}$$

Proof. An involution is determined by its (± 1) -eigenlattices, and these are mutual annihilators, so the map is injective; it is surjective by the 2-torsion remark above. If h commutes with ε it preserves each eigenlattice. Conversely if h preserves $N = \Lambda_-(\varepsilon)$ it preserves $\text{ann}_\Lambda(N) = \Lambda_+(\varepsilon)$, and ε is ∓ 1 on these, so $h\varepsilon h^{-1} = \varepsilon$. $\square$

Thus every statement below is a statement about invariant EE_8 -sublattices.

3. TWO MODELS OF Λ , EXPLICITLY IDENTIFIED

We use two constructions. The *binary model* builds Λ from the extended binary Golay code $\mathcal{G}_{24}$ in the standard way; its Gram matrix is G_{24} . The *ternary model* builds Λ from the extended ternary Golay code $\mathcal{G}_{12}$ over $\mathbb{Z}[\omega]$, passing through the Niemeier lattice A_2^{12} and a two-dimensional isotropic glue in $\mathbb{F}_3^4$; its Gram matrix is G_t . In the ternary model, multiplication by ω is an isometry.

Rather than invoke Conway’s characterisation of Λ as the unique even unimodular rootless lattice of rank 24, we identify the two models by a matrix.

Theorem 2 (`leech_models_isometric`). *There is $T \in GL_{24}(\mathbb{Z})$ with $T G_{24} T^t = G_t$ and $\det T = \pm 1$.*

The matrix was found by taking the 48 norm-8 vectors in a single class modulo 2Λ — these form a coordinate frame — reading the ternary model in frame coordinates, matching the resulting 759 octads to the binary Golay octads by a Steiner-system isomorphism, and solving for signs. Consequently the two theorems below concern one lattice, not two lattices known to be isomorphic.

4. THE NEGATIVE HALF

Let π_1 be the permutation $x \mapsto (x+1)/(-x)$ of $\mathbb{P}^1(\mathbb{F}_{23})$, a fixed-point-free element of order 3 in $PSL_2(23) \subset M_{24}$, and let $\sigma_1 \in \text{Co}_0$ be the induced isometry of the binary model. Write $R = \text{Fix}(\sigma_1)$ and $Q = \{v \in \Lambda : v + \sigma_1 v + \sigma_1^2 v = 0\}$.

Theorem 3. *R has rank 8 with $\frac{1}{3} \cdot \text{Gram}(R)$ even unimodular, Q has rank 16 and is isometric to $A_2 \otimes E_8$, $\Lambda/(Q \oplus R) \cong \mathbb{F}_3^8$, and*

no EE_8 -involution of Co_0 commutes with σ_1 .

Proof sketch. By Proposition 1 we must show Λ contains no σ_1 -invariant EE_8 -sublattice N . On Q the element σ_1 satisfies $x^2 + x + 1 = 0$, so Q is a $\mathbb{Z}[\omega]$ -module and σ_1 has no fixed vectors there; on R it is the identity. The certificates `sqrt3E8_certificate`, `isometry_Q_A2E8` and the Smith form of $[\Lambda : Q \oplus R]$ give the first three assertions.

Suppose N is σ_1 -invariant. Then $\sigma_1|_N$ lies in $\text{Aut}(N) \cong W(E_8)$ and has order 1 or 3. If order 1, then $N \subseteq R$; both have rank 8, so $N = R$, impossible since $\min N = 4$ while $\min R = 6$. So $\sigma_1|_N$ has order 3, with fixed rank $f \in \{0, 2, 4, 6\}$ (even, by the $\mathbb{Z}[\omega]$ -structure on N/N^{σ_1}).

Case $f > 0$. Then $N^{\sigma_1} \subseteq R$, so $\min N^{\sigma_1} \geq 6$. By Carter’s classification, an order-3 element of $W(E_8)$ lies in a class A_2^k , $k = 1, 2, 3$; for each we exhibit an explicit representative together with a norm-2 root of E_8 — equivalently a norm-4 vector of $\sqrt{2}E_8$ — in its fixed lattice (`Ak_root_in_fix`), with the fixed rank pinned exactly by an adjugate rank certificate. This contradicts $\min \geq 6$.

Case $f = 0$. Then $N \subseteq Q$ and N is a $\mathbb{Z}[\omega]$ -submodule. The 720 minimal vectors of Q fall into 120 ω -hexagons; those of N span 40 of them, and any two hexagons meeting N have all cross inner products even. Since ω acts irreducibly on each hexagon plane, N is the saturation of the $\mathbb{Z}$ -span of a compatible 4-clique of hexagons. Every such span has determinant $2^8 \cdot 3^4$, whereas $\det N = 2^8$; hence the saturation index is exactly 9, so 3 divides it. But the 8×16 span matrix of every one of the 122 850 compatible 4-cliques has *full rank modulo 3* (`no_index9_saturation`), so its saturation index is prime to 3. Contradiction. $\square$

The $f = 0$ sweep is an exhaustive finite computation carried out inside the Lean kernel’s compiled evaluator; the case analysis and the reduction to it are by hand.

5. THE POSITIVE HALF

Work in the ternary model and let σ_2 be multiplication by ω .

Theorem 4. σ_2 is an isometry of Λ with $\sigma_2^2 + \sigma_2 + 1 = 0$, hence fixed-point-free, of trace -12 . Let ε be the involution induced by the permutation

$$(2\ 6)(3\ 7)(4\ 10)(8\ 9)$$

of the twelve $\mathbb{Z}[\omega]$ -coordinates. Then ε is an EE₈-involution, $[\varepsilon, \sigma_2] = 1$, and

$$g := \varepsilon \sigma_2$$

has order 6, with $g^3 = \varepsilon$, $g^4 = \sigma_2$, $\text{tr } g = -4$, and characteristic polynomial $\chi_g = \Phi_6^4 \Phi_3^8$. In particular g is fixed-point-free.

Proof. All assertions are identities between explicit integer matrices and are certified by `sigma_fpf`, `eps_trace8`, `minus_lattice_is_sqrt2E8`, `eps_commutes_sigma`, `g_order6`, `g_powers` and `g_cyclotomic`, the last being $(g^2 - g + 1)(g^2 + g + 1) = 0$. Commutation is automatic: ε is $\mathbb{Z}[\omega]$ -linear. $\square$

Combining Theorems 2, 3 and 4: a single lattice carries an order-3 isometry with no commuting EE₈-involution, and another with one.

Remark 5 (How many). Within the sign-monomial subgroup of the $\mathbb{Z}[\omega]$ -linear automorphisms of Λ there are exactly 55 EE₈-involutions commuting with σ_2 ; allowing the full unit group $\{\pm 1, \pm \omega, \pm \omega^2\}$ on the twelve coordinates, exactly 1485, pairwise distinct as involutions, all of trace 8. These counts are computational and are not among the formalised statements; they are lower bounds for the count in $C_{\text{Co}_0}(\sigma_2)$.

Remark 6 (Two glues, one σ_1). Λ contains $(\sqrt{2}E_8)^3$ with glue group $\mathbb{F}_2^{12}$ (Turyn), and $Q \perp R$ with glue group $\mathbb{F}_3^8$. The isometry σ_1 permutes the three $\sqrt{2}E_8$ blocks cyclically and acts on the binary glue as the corresponding cyclic shift; on the ternary glue it acts *trivially*, because $\omega - 1$ is an associate of $\theta = 1 - \omega$. The 2-local and 3-local data of Λ thus coexist without being fused by σ_1 ; Theorem 3 is the precise form of that statement, and Theorem 4 shows the obstruction vanishes at the fixed-point-free class.

6. THE MECHANISM: $3 \nmid 8$

Both halves are explained by a fact about the number 8.

Theorem 7. Let p be a fixed-point-free permutation of order 3 of a 24-element set. Then no 8-element subset is p -invariant. Consequently π_1 stabilises no Golay octad.

By contrast, each of the three order-4 conjugacy classes of M_{24} — cycle types $1^4 2^2 4^4$, $2^4 4^4$ and 4^6 — stabilises an octad; the numbers of stabilised octads are 11, 7 and 3 respectively.

Proof. A p -invariant set is a union of p -orbits. If p is fixed-point-free of order 3, every orbit has size 3, so every invariant set has cardinality divisible by 3; and $3 \nmid 8$. The second statement is a finite check (`p4A/B/C.invariant_octad_count`); the orbit sizes there are 1, 2, 4, all divisors of 8. $\square$

The relevance is the following construction. If O is an octad, then $M_O = \Lambda \cap \mathbb{R}^O$ is a Turyn block, isometric to $\sqrt{2}E_8$. A monomial isometry — a signed permutation — preserves every coordinate subspace determined by an invariant set. Hence:

Proposition 8. Let $h \in \text{Co}_0$ be monomial of order 4. Then h commutes with an EE₈-involution.

Proof. The permutation part of h has order 1, 2 or 4; order 1 would make h an involution. Every class of M_{24} of order 2 or 4 stabilises an octad O (Theorem 7 for order 4; for the classes 2A, 2B the counts are 71 and 15). Then M_O is h -invariant and $\sqrt{2}E_8$, and Proposition 1 supplies ε . An explicit instance for the class 4⁶ is certified: $\text{Fix}(h)$ has rank 6, $M_O \cong \sqrt{2}E_8$, h preserves M_O , and the resulting ε has trace 8 and commutes with h (`eps_commutes_with_g`). $\square$

So the free construction is available at every even order and blocked at 3 for an arithmetic reason. That is exactly why Theorem 3 required an exhaustive sweep: the EE_8 -sublattices of Λ are not all Turyn blocks, and once the octad route is closed nothing cheaper survives.

Remark 9. Proposition 8 covers only monomial elements; the monomial subgroup $2^{12}:M_{24}$ has index roughly 8.3×10^6 in Co_0 . We conjecture that *every* order-4 isometry of Λ with nonzero fixed lattice commutes with an EE_8 -involution; equivalently, that the permutation character of Co_0 on EE_8 -sublattices does not vanish on any such class. Theorem 3 says that character *does* vanish on the class of σ_1 . The classes of M_{24} stabilising no octad are those of cycle type 3^8 , 6^4 , $2^2 10^2$, $1^2 11^2$, 12^2 , $3 \cdot 21$ and $1 \cdot 23$; none has order 4, and each is a candidate for a further negative result.

7. FORMALISATION

The finite verifications are carried out in Lean 4 (core only, no Mathlib). All proofs are by `native_decide` on explicit integer data; unimodularity is certified by exhibiting an integer inverse, so no determinant routine is needed.

File	Content	Thms
<code>E8TrialityCert.lean</code>	Leech Gram; $\sqrt{2}E_8$ blocks; 2^{12} glue; $2 \cdot A_9 \subset W(E_8)$	20
<code>E8TrialityCertII.lean</code>	ternary 3^8 glue; σ_1 -triviality; Smith form; solid	12
<code>E8TrialityCertIII.lean</code>	$2 \cdot \tilde{S}_9 \subset \text{Aut}(E_8 \perp E_8)$; Milnor pair	12
<code>E8TrialityCertIV.lean</code>	Theorem 3: the 122 850-clique sweep; A_2^k roots	20
<code>E8TrialityCertV.lean</code>	Theorem 4; Theorem 2	18
<code>E8TrialityCertVI.lean</code>	Theorem 7; Proposition 8 witness	26
total, no sorry		108

Trust base. Because every proof uses `native_decide`, the theorems depend on the axioms `propext`, `Lean.ofReduceBool` and `Lean.trustCompiler`. The latter two extend trust to the Lean compiler’s evaluation of compiled code; they are *not* kernel-standard. The heaviest check, `no_sigma_invariant_eight_set`, sweeps all 2^{24} subsets of the 24 points.

Unformalised inputs. (i) Uniqueness of E_8 in its genus, which licenses the label $\sqrt{2}E_8$ (and $\sqrt{3}E_8$) wherever it appears; (ii) Carter’s classification of the order-3 classes of $W(E_8)$ as A_2^k ; (iii) the reduction of Theorem 3 to the case split on f ; (iv) the Schur multiplier of A_9 , used only in `E8TrialityCertIII`; (v) the passage from “permutation stabilises an octad” to “monomial isometry preserves the block”, and the exhaustiveness of the M_{24} class list in Proposition 8. Conway’s uniqueness theorem is *not* used, by Theorem 2.

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REFERENCES

- [CS] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Springer, 1999.
- [Ca] R. W. Carter, *Conjugacy classes in the Weyl group*, Compositio Math. **25** (1972), 1–59.
- [GL1] R. L. Griess Jr. and C. H. Lam, *EE_8 -lattices and dihedral groups*, Pure Appl. Math. Q. **7** (2011).
- [GL2] R. L. Griess Jr. and C. H. Lam, *A moonshine path from E_8 to the Monster*, J. Pure Appl. Algebra **215** (2011); arXiv:0910.2057.
- [L] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE 28 (2021).